

direction of the arrow (or in either direction for neutral particle lines without arrows.)

(ii) For each vertex of type i , include a factor

$$-i(2\pi)^4 g_i \delta^4 \left(\sum p + \sum q - \sum p' - \sum q' \right) \quad (6.3.2)$$

with the momentum sums having the same meaning as in (6.3.1). This delta function ensures that the four-momentum is conserved at every point in the diagram. For each external line running upwards out of the diagram, include a factor $(2\pi)^{-3/2} u_\ell^*(\mathbf{p}'\sigma'n')$ or $(2\pi)^{-3/2} v_\ell(\mathbf{p}'\sigma'n')$, for arrows pointing upwards or downwards, respectively. For each external line running from below into the diagram, include a factor $(2\pi)^{-3/2} u_\ell(\mathbf{p}\sigma n)$ or $(2\pi)^{-3/2} v_\ell^*(\mathbf{p}\sigma n)$, for arrows pointing upwards or downwards, respectively. For each internal line with ends labelled ℓ and m , the arrow pointing from m to ℓ , and carrying a momentum label q^μ , include as a factor the integrand of the integral for $-i\Delta_{\ell m}(q)$:

$$-i(2\pi)^{-4} P_{\ell m}(q) / (q^2 + m_\ell^2 - i\epsilon). \quad (6.3.3)$$

A reminder: for scalars or antiscalars of four-momentum q , the u s and v s are simply $(2q^0)^{-1/2}$, while the polynomial $P(q)$ is unity. For Dirac spinors of four-momentum p and mass M , the u s and v s are the normalized Dirac spinors described in Section 5.5, and the polynomial $P(p)$ is the matrix $(-i\gamma_\mu p^\mu + M)\beta$.

(iii) Integrate the product of all these factors over the four-momenta carried by internal lines, and sum over all field indices ℓ, m , etc.

(iv) Add up the results obtained in this way from each Feynman diagram.

Additional combinatoric factors and fermionic signs may need to be included, as described in parts (v) and (vi) of Section 6.1. Examples will be given at the end of this section.

We have a four-momentum integration variable for every internal line, but many of these are eliminated by the delta functions associated with vertices. Since energy and momentum are separately conserved for each connected part of a Feynman diagram, there will be C delta functions left over in a graph with C connected parts. Hence in a diagram with I internal lines and V vertices, the number of independent four-momenta that are *not* fixed by the delta functions is $I - [V - C]$. This is clearly also the number L of independent loops:

$$L = I - V + C, \quad (6.3.4)$$

which is defined as the maximum number of internal lines that can be cut without disconnecting the diagram, because any such and only